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QUARTERLY EXAMINATION -2025

CLASS:12 **MATHEMATICS** **Reg.No**

Time : 3.00 Hours **MARKS : 90**

PART – I

Answer all the questions . **20 x 1 = 20**

1. If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then $|\text{adj}(AB)| =$ ----------

1) – 40 2) – 80 3) – 60 4) – 20

2. If $A^T A^{-1}$ is symmetric, then A^2 is ----------

1) A^{-1} 2) $(A^T)^2$ 3) $(A)^T$ 4) $(A^{-1})^2$

3. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$ is ----------

1) 1 2) 2 3) 4 4) 3

4. The product of all four values of $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is

(1)-2 (2)-1 (3)1 (4)2

5. The principal argument of $(\sin 40^0 + i\cos 40^0)^5$ is

(1)-110^0 (2) -70^0 (3) 70^0 (4) 110^0

6. If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is

(1)-2 (2)-1 (3)1 (4)2

7. If $x^3 + 12x^2 + 10ax + 1999$ definitely has a positive root, if and only if

(1)$a \geq 0$ (2) $a > 0$ (3) $a < 0$ (4) $a \leq 0$

8. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if k satisfies

(1)$|k| \leq 6$ (2) $k = 0$ (3)$|k| > 6$ (4)$|k| \geq 6$

9. If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$; *then* $\cos^{-1} x + \cos^{-1} y$ is equal to .....

1) $\frac{2\pi}{3}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) π

10. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is.....

1) [1,2] 2) [−1, 1] 3) [0, 1] 4) [−1, 0]

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11. $\sin^{-1}(2\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x) =$.....

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

12. If $\sin^{-1}\frac{x}{5} + \text{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is .....

1) 4 2) 5 3) 2 4) 3

13. The radius of the circle passing through the point (6,2) two of whose diameter are $x + y = 6$ and $x + 2y = 4$ is

(1) 10 (2) $2\sqrt{5}$ (3) 6 (4) 4

14. If $x + y = K$ is a normal to the parabola $y^2 = 12x$, then the value of k is

(1) 3 (2) -1 (3) 1 (4) 9

15. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

(1) 2ab (2) ab (3) $\sqrt{ab}$ (4) $\frac{a}{b}$

16. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (11,2) the coordinates of the other end are

(1) (-5,2) (2) (3,-5) (3) (5,−2) (4) (−2,5)

17. The angle between the line $\vec{r} = (\vec{i} + 2\vec{j} - 3\vec{k}) + t(2\vec{i} + \vec{j} - 2\vec{k})$ and the plane $\vec{r}.(\vec{i} + \vec{j}) + 4 = 0$ is

(1) 0^0 (2) 30^0 (3) 45^0 (4) 60^0

18. The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$ is

(1) $\frac{\sqrt{7}}{2\sqrt{2}}$ (2) $\frac{7}{2}$ (3) $\frac{\sqrt{7}}{2}$ (4) $\frac{7}{2\sqrt{2}}$

19. Volume of the parallelepiped whose coterminous edges are represented by the vectors $-6\hat{i} + 14\hat{j} + 10\hat{k}$, $14\hat{i} - 10\hat{j} - 6\hat{k}$ and $2\hat{i} + 4\hat{j} - 2\hat{k}$ is

1) 700 2) 500 3) 720 4) 520

20. The type of the conic $x^2 + y^2 + x - y = 0$ is

1) parabola 2) ellipse 3) hyperbola 4) circle

PART – II

Answer any seven questions. Question No.30 is compulsory. **7 x 2 = 14**

21. If A is symmetric, prove that adj A is also symmetric.

22. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

M.Poovarasan M.Sc B.Ed
PG Asst in Chemistry
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23. Show that (i) $(2+i\sqrt{3})^{10}-(2-i\sqrt{3})^{10}$ is purely imaginary
24. Find the modulus and principal argument of $\sqrt{3}-i$.
25. If α and β are the roots of the quadratic equation $17x^2+43x-73=0$, construct a quadratic equation whose roots are $\alpha+2$ and $\beta+2$.
26. Find the principal value of $\cot^{-1}(\sqrt{3})$.
27. Find the equation of the ellipse with foci $(\pm 2,0)$, vertices $(\pm 3,0)$.
28. If $\vec{a}=\hat{\imath}-2\hat{\jmath}+3\hat{k}, \vec{b}=2\hat{\imath}+\hat{\jmath}-2\hat{k}, \vec{c}=3\hat{\imath}+2\hat{\jmath}+\hat{k}$, find $(\vec{a}\times\vec{b})\times\vec{c}$
29. Show that the lines $\frac{x-1}{4}=\frac{2-y}{6}=\frac{z-4}{12}$ and $\frac{x-3}{-2}=\frac{y-3}{3}=\frac{5-z}{6}$ are parallel.
30. Solve: $x^4-25x^2+144=0$

PART – III

Answer any Seven questions. Question No.40 is compulsory. **7 x 3 = 21**

31. Solve the following system of linear equations by matrix inversion method:
$2x+5y=-2, x+2y=-3$.
32. Find the rank of $\begin{bmatrix}1 & 1 & 1 & 3\\ 2 & -1 & 3 & 4\\ 5 & -1 & 7 & 11\end{bmatrix}$.
33. Simplify $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3$. into rectangular form.
34. Find the square root of $6-8i$.
35. Find a polynomial equation of minimum degree with rational coefficients, having $2i+3$ as a root.
36. Find the domain of $\sin^{-1}(2-3x^2)$.
37. If the equation $3x^2+(3-p)xy+qy^2-2px=8pq$ represents a circle, find p and q. Also determine the centre and radius of the circle.
38. Prove that $[\vec{a}-\vec{b},\vec{b}-\vec{c},\vec{c}-\vec{a}]=0$.
39. Find the coordinates of the point where the straight line $\vec{r}=(2\hat{\imath}-\hat{\jmath}+2\hat{k})+t(3\hat{\imath}+4\hat{\jmath}+2\hat{k})$ intersects the plane $x-y+z-5=0$.
40. Prove that $\cos^{-1}x=\pi-\sin^{-1}\sqrt{1-x^2}$, $x<0$

PART – IV

Answer all the questions. **7 x 5 = 35**

41. **(a)** Investigate the values of λ and μ the system of linear equations
$2x+3y+5z=9, 7x+3y-5z=8, 2x+3y+\lambda z=\mu$, have
(i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(OR)

(b) Solve the following system of linear equations by Cramer's rule:
$\frac{3}{x}-\frac{4}{y}-\frac{2}{z}-1=0, \frac{1}{x}+\frac{2}{y}+\frac{1}{z}-2=0, \frac{2}{x}-\frac{5}{y}-\frac{4}{z}+1=0$

42. **(a)** If $2\cos\alpha=x+\frac{1}{x}$ and $2\cos\beta=y+\frac{1}{y}$ show that
(i) $xy-\frac{1}{xy}=2i\sin(\alpha+\beta)$ (ii) $\frac{x^m}{y^n}-\frac{y^n}{x^m}=2i\sin(m\alpha-n\beta)$

(OR)

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(*b*) If z_1, z_2 and z_3 are three complex numbers such that $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$ show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$.

43. (*a*) Solve : (i) $(x-5)(x-7)(x+6)(x+4) = 504$

(OR)

(*b*) Solve the following equation: $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$.

44. (*a*) Solve: $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

(OR)

(*b*) Find the value of $\cot^{-1}(1) + \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) - \sec^{-1}(-\sqrt{2})$

45. (*a*) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of $4m$ when it is $6m$ away from the point of projection. Finally it reaches the ground $12m$ away from the starting point. Find the angle of projection.

(OR)

(*b*) Find the centre, foci, and eccentricity of the hyperbola

$11x^2 - 25y^2 - 44x + 50y - 256 = 0$.

46. (*a*) Find the equation of the circle passing through the points (1,1),(2,-1) and (3,2).

(OR)

(*b*) Find the image of a point (5, 4, 2) in the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$.

47. (*a*) Using vector method, prove that $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$.

(OR)

(*b*) Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} - \hat{j} + 4\hat{k})$ and perpendicular to plane $\vec{r}.(\hat{i} + 2\hat{j} + \hat{k}) = 8$.

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